

Deep Learning Systems Analysis Report

AI Job Market Persona Discovery - Transformer vs Autoencoder

Nina Haide | May 10th, 2026

1. Report Overview

This report presents a deep learning analysis aimed at discovering AI job market personas from job description text using unsupervised clustering on neural text representations. The dataset is a custom collection of 1,328 AI job postings across six European countries, collected via the Adzuna Jobs API and enriched with full descriptions fetched from source pages. Two models are compared: a PyTorch autoencoder trained on TF-IDF vectors as the baseline and a pre-trained DistilBERT Transformer used in inference mode as the experimental model, with k-means clustering applied to the representations of both. This project directly extends the keyword-based skill clustering of the previous project, replacing hand-crafted features with learned and pre-trained neural representations.

2. Dataset and Task Description

The dataset was created using a custom two-script pipeline: `fetch_ai_jobs_personas.py` collected job postings via the Adzuna Jobs API across Germany, United Kingdom, France, Netherlands, Austria and Switzerland and `enrich_descriptions.py` fetched full job description text from source pages to overcome the API's 500-character truncation limit. After filtering postings where enrichment failed, the modeling dataset comprises 1,088 job postings with full descriptions, stored in `adzuna_ai_jobs_europe_enriched.csv` across 19 columns. Key fields include `description_full` (the text input to both models), `role_category` (a metadata label used for post-clustering evaluation), `country` and `title`. The task is unsupervised - there is no target variable. The goal is to learn a representation of each job description and cluster those representations into meaningful role archetypes, referred to as personas, that reflect the semantic structure of the European AI job market rather than the labels employers assign. The United Kingdom contributed the largest share of postings (675), followed by Germany (409), France (137), Switzerland (43), Netherlands (34) and Austria (30).

3. Model Architecture and Design Decisions

Baseline - PyTorch Autoencoder on TF-IDF vectors

Job descriptions were first vectorised using TF-IDF (Sparck Jones, 1972) with a vocabulary of 500 terms, producing a $1,088 \times 500$ sparse feature matrix. An autoencoder was then trained to compress these vectors into a 32-dimensional latent representation and reconstruct them. The encoder follows a $500 \rightarrow 128 \rightarrow 32$ architecture with ReLU activations; the decoder mirrors it at $32 \rightarrow 128 \rightarrow 500$ with a Sigmoid output, appropriate since TF-IDF values fall in $[0, 1]$. The model was trained for 50 epochs using Mean Squared Error loss and the Adam optimiser ($\text{lr}=0.001$) with a batch size of 32.

An autoencoder was selected as the baseline because it satisfies the PyTorch training requirement while providing a principled neural compression of the input space. The 32-dimensional latent space is small enough to force meaningful compression, and the reconstruction objective ensures the latent vectors retain the most informative structure from the original TF-IDF representation. As Goodfellow et al. (2016) describe, autoencoders learn efficient codings by minimising reconstruction error, making the latent space a compact summary of input structure.

Experimental - DistilBERT embeddings

The pre-trained DistilBERT model (Sanh et al., 2019) was loaded in inference mode. Each job description was tokenised and encoded, with the [CLS] token representation taken as the document embedding - a standard practice for sentence-level tasks using BERT-family models (Devlin et al., 2019). This produced a $1,088 \times 768$ embedding matrix requiring no training. K-means clustering ($k=5$, $n_{\text{init}}=10$) was applied identically to both model outputs to ensure the comparison isolates the representation method alone.

4. Experimental Comparison

The single change between baseline and experimental model is the text representation method: TF-IDF compressed through a trained autoencoder versus pre-trained DistilBERT embeddings used directly. TF-IDF encodes word frequency relative to the corpus but has no understanding of word meaning or context - two descriptions using different words to describe the same role will appear distant in TF-IDF space. DistilBERT, trained on a large text corpus using masked language modelling, encodes contextual meaning such that semantically similar descriptions map to nearby points in embedding space regardless of exact vocabulary (Sanh et al., 2019).

This comparison is meaningful because it directly tests the core claim of deep language models: that pre-trained contextual representations produce more useful features for

downstream tasks than frequency-based methods. For job market persona discovery specifically, semantic understanding matters - a role described as 'building production ML pipelines' and one requiring 'end-to-end machine learning systems' should cluster together, which TF-IDF may not achieve but DistilBERT is designed to handle.

5. Results and Interpretation

Both models were evaluated using the silhouette score (Rousseeuw, 1987), which measures cluster cohesion and separation on a scale from -1 to +1. The baseline autoencoder and the DistilBERT experimental model were both clustered at $k=5$ using identical k-means configurations. Silhouette scores were also computed across $k = 2$ to $k = 8$ to assess whether the choice of k affects the relative performance of the two models.

The cluster composition cross-tabulation - comparing cluster assignments against the human-assigned `role_category` labels - provides qualitative evidence of which model produces more coherent role archetypes. Stronger alignment with role categories in the experimental model would support the hypothesis that semantic representations better capture the true structure of the AI job market. PCA projections of both models (`viz_cluster_comparison.png`) provide visual evidence of cluster separation, with more compact and distinct clusters indicating a more informative representation.

6. Limitations and Risks

Dataset limitations

The United Kingdom contributes 51% of postings, meaning the discovered personas primarily reflect the British AI job market. Austria (30) and Netherlands (34) are too underrepresented to contribute meaningfully to cluster formation. Additionally, 240 of 1,328 postings (18%) could not be enriched due to JavaScript-rendered employer pages - these were excluded, introducing a potential selection bias toward companies using simpler web infrastructure.

Architecture limitations

K-means assumes spherical clusters and equal variance, which is unlikely to hold in a high-dimensional semantic embedding space. DistilBERT's maximum input length of 512 tokens means descriptions longer than approximately 380 words are truncated, potentially losing key skill requirements that appear at the end of postings. The [CLS] token embedding, while standard, may not be the optimal sentence

representation for domain-specific clustering - mean pooling of all token embeddings is an alternative worth exploring (Reimers & Gurevych, 2019).

7. Ethical and Responsible Use

The most significant ethical concern is the potential misuse of persona labels in hiring or screening contexts. If the cluster archetypes discovered in this project were applied to filter or rank job applicants - mapping CVs to personas and preferencing certain clusters - the system would encode and amplify any biases present in the training data. Since the dataset is dominated by UK and German job postings, the personas reflect those markets' norms and may systematically disadvantage candidates from different educational backgrounds, non-English-speaking countries or career paths not well-represented in the Adzuna index. As Barocas and Hardt (2023) note, unsupervised models trained on biased data can produce clusters that encode protected characteristics without explicitly modelling them.

To mitigate this risk, the project scope is explicitly limited to descriptive market analysis - understanding what roles exist - rather than prescriptive screening. Cluster labels are assigned based on job description content, not applicant characteristics and no individual person's data is used at any stage. Any extension of this work to CV analysis or candidate matching would require a dedicated bias audit and compliance review under the EU AI Act, which classifies recruitment AI systems as high-risk applications.

8. Future Improvements

The most impactful near-term improvement would be replacing K-Means with HDBSCAN (Campello et al., 2013), a density-based clustering algorithm that does not require k to be specified in advance and handles non-spherical cluster shapes more effectively. This is particularly relevant for DistilBERT embeddings, where cluster boundaries are unlikely to be spherical in the original 768-dimensional space.

A second improvement would be to use Sentence-BERT (Reimers & Gurevych, 2019) instead of vanilla DistilBERT - a model specifically fine-tuned to produce semantically meaningful sentence embeddings rather than document-level [CLS] tokens. Fine-tuning on a domain-specific corpus of job descriptions would further improve the quality of representations for this specific task.

On the data side, extending the collection to include job postings from additional European countries and sources beyond Adzuna would reduce the UK dominance and produce more geographically representative personas. Collecting postings over multiple time periods would also enable temporal analysis of how AI role archetypes evolve as the technology matures.

9. References

- Barocas, S., & Hardt, M. (2023). Fairness and machine learning: Limitations and opportunities. MIT Press. <https://fairmlbook.org>
- Campello, R. J. G. B., Moulavi, D., & Sander, J. (2013). Density-based clustering based on hierarchical density estimates. In *Advances in Knowledge Discovery and Data Mining* (pp. 160–172). Springer. https://doi.org/10.1007/978-3-642-37456-2_14
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional Transformers for language understanding. In *Proceedings of NAACL-HLT 2019* (pp. 4171–4186). <https://doi.org/10.18653/v1/N19-1423>
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep learning. MIT Press. <https://www.deeplearningbook.org>
- Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of EMNLP-IJCNLP 2019* (pp. 3982–3992). <https://doi.org/10.18653/v1/D19-1410>
- Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65. [https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7)
- Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter. arXiv preprint arXiv:1910.01108. <https://arxiv.org/abs/1910.01108>
- Sparck Jones, K. (1972). A statistical interpretation of term specificity and its application in retrieval. *Journal of Documentation*, 28(1), 11–21. <https://doi.org/10.1108/eb026526>
- Adzuna. (2026). Adzuna Jobs API [Data collection API]. <https://developer.adzuna.com>